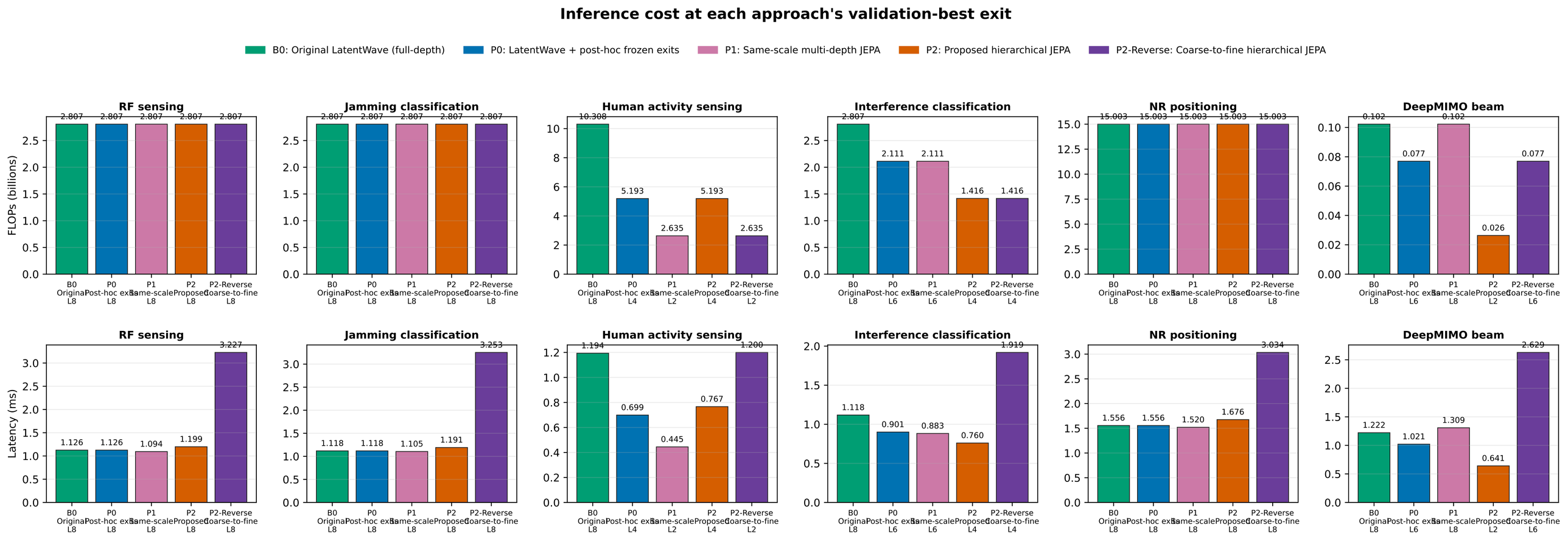
Inference cost at each approach's validation-best exit
B0: Original LatentWave (full-depth)
P0: LatentWave + post-hoc frozen exits
P1: Same-scale multi-depth JEPA
P2: Proposed hierarchical JEPA
P2-Reverse: Coarse-to-fine hierarchical JEPA
RF sensing
Jamming classification
Human activity sensing
Interference classification
NR positioning
DeepMIMO beam
FLOPs (billions)
Latency (ms)
B0 Original L8
P0 Post-hoc exits L8
P0 Post-hoc exits L4
P0 Post-hoc exits L6
P0 Post-hoc exits L2
P1 Same-scale L8
P1 Same-scale L2
P1 Same-scale L6
P2 Proposed L8
P2 Proposed L4
P2 Proposed L2
P2-Reverse Coarse-to-fine L8
P2-Reverse Coarse-to-fine L2
P2-Reverse Coarse-to-fine L4
P2-Reverse Coarse-to-fine L6
2.807
2.807
2.807
2.807
2.807
10.308
5.193
2.635
5.193
2.635
2.807
2.111
2.111
1.416
1.416
15.003
15.003
15.003
15.003
15.003
0.102
0.077
0.102
0.026
0.077
1.126
1.126
1.094
1.199
3.227
1.118
1.118
1.105
1.191
3.253
1.194
0.699
0.445
0.767
1.200
1.118
0.901
0.883
0.760
1.919
1.556
1.556
1.520
1.676
3.034
1.222
1.021
1.309
0.641
2.629
Batch-1 synchronized CUDA timing on the RTX 5070 Ti; lower is better. Layer labels are printed below each method.